magnetic bearings depend on the vibration frequency, whereas for oil bearings they depend on the rotating speed. Compared to oil bearings, magnetic bearings have a lower stiffness. In the frequency range of 50 to 200 Hz, the stiffness of the magnetic bearings is only about one sixth of the stiffness that an oil bearing for Mopico would have at a speed of 10,000 rpm. In order to prevent large static deflections due to the low stiffness, the controller of the magnetic bearing has an integration term.

Phase lead cells in the controller provide the damping of the magnetic bearing—however, only in a limited frequency range. At very low frequencies (below about 30 Hz) and at very high frequencies (above 1500 Hz), the damping is negative. In the frequency range of 50 to 200 Hz, the damping coefficient of the magnetic bearing is of the same order of magnitude as that of an oil bearing.

The theoretical model of the rotor bearing system is well proven by measured closed loop transfer functions (relation of the displacement at the sensor and a magnetic excitation force at the bearing).

The rotor can be safely run up to the design maximum speed of 10,000 rpm. The vibration level at this speed is not more than about 50 percent of the limit due to amplifier saturation. If the cold rotor is run up slowly to this speed (within 10 to 15 min), the level is even lower.

Isotherm turbocompressors

Turbocompressors with the lowest power consumption. The word *isotherm* describes the principal feature of these machines; the flow medium is cooled intensively during the compression process in order to come as close as possible to ideal isothermal compression, giving maximum efficiency and therefore minimum power requirement.

Isotherm compressors are particularly suited to oil-free compression of air, oxygen, or nitrogen to discharge pressure up to 13 bar. For higher pressure ratios a booster may be added to the compressor train. The isotherm compressors are widely used in air separation plants, fertilizer plants, iron and steelworks, and for compressed-air supplies to mines. See Figs. C-192 and C-193 for illustration of some typical applications. The design has been steadily improved since they were first introduced in 1913. An important factor for their great success is their low specific power requirement, a result of flow path and intensive intercooling. Compact, standardized construction and high availability are required of them. More than 1000 isotherm compressors are in service throughout the world, many with up to 200,000 h to their credit and a time between overhauls of three to six years.

Operating range. See Fig. C-193.

Design features. Design features are incorporated to suit an end user's application. Typically they include:

1. Low power consumption
2. Resistance against corrosion
3. High rotor stability = low vibration level
4. Low noise level
5. Shaft-string configuration
6. Ease of installation
7. Simple maintenance programs
8. Minimized space requirement
9. High reliability

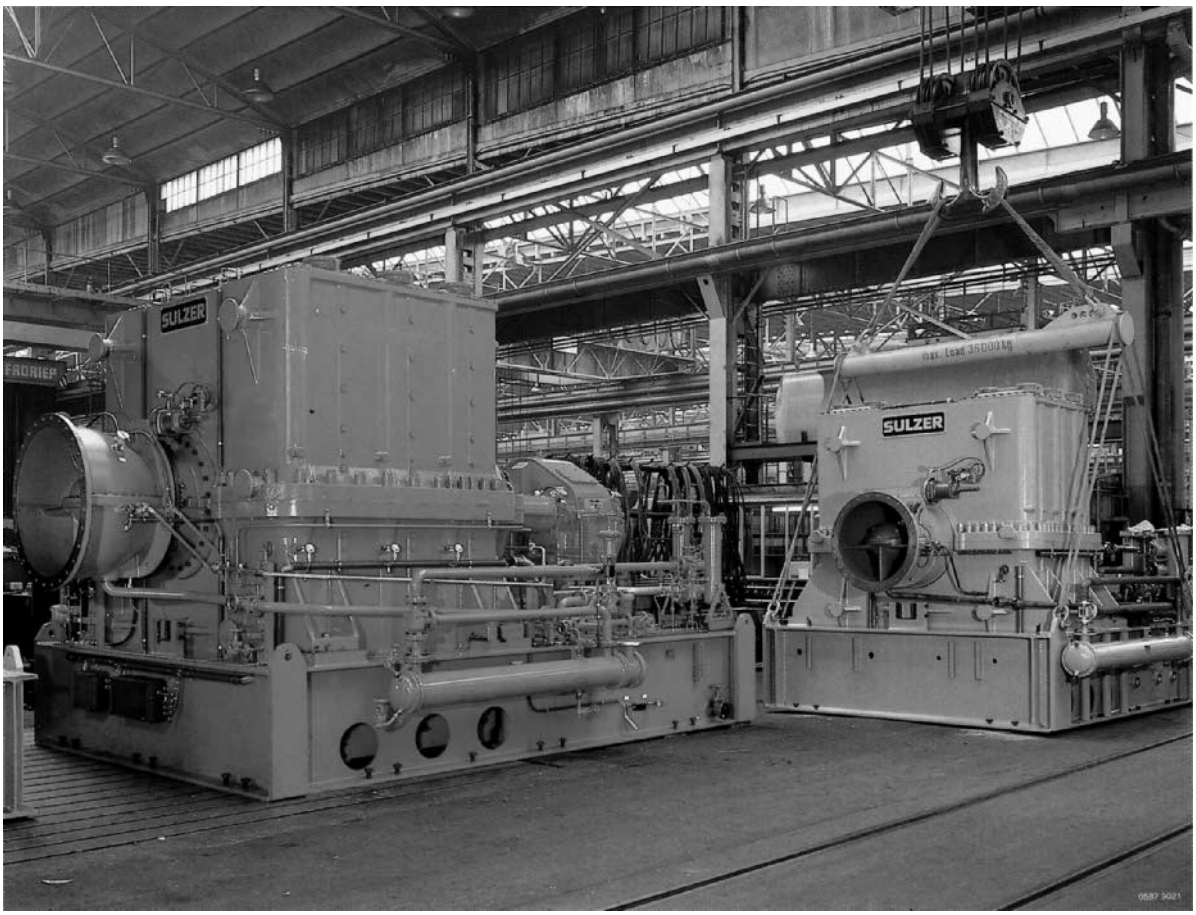
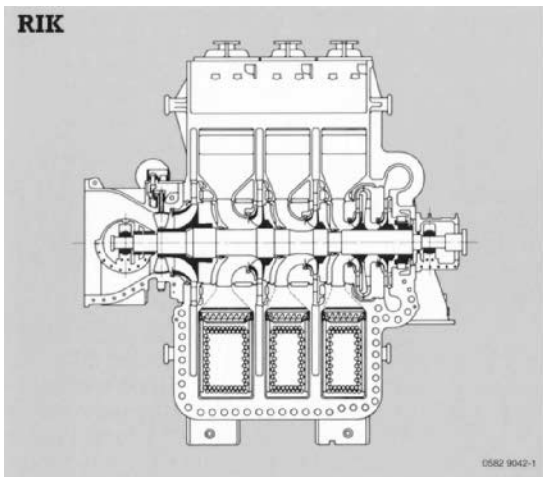
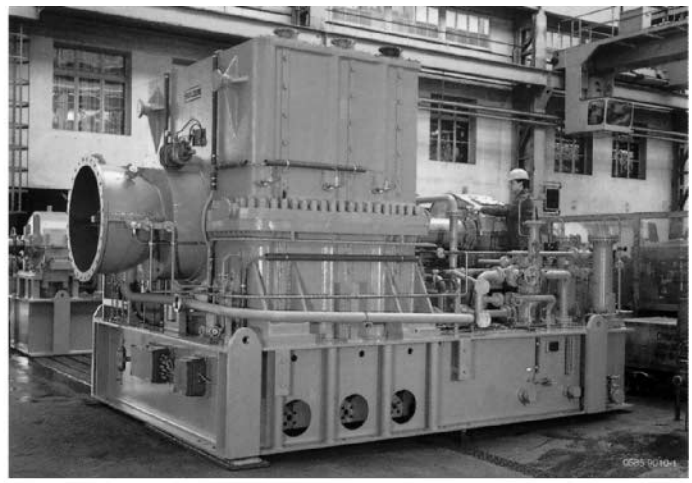


FIG. C-192 Air compressors, type RIK 80 and RIK 56. Transportation as a single-lift package. (Source: Sulzer-Burckhardt.)

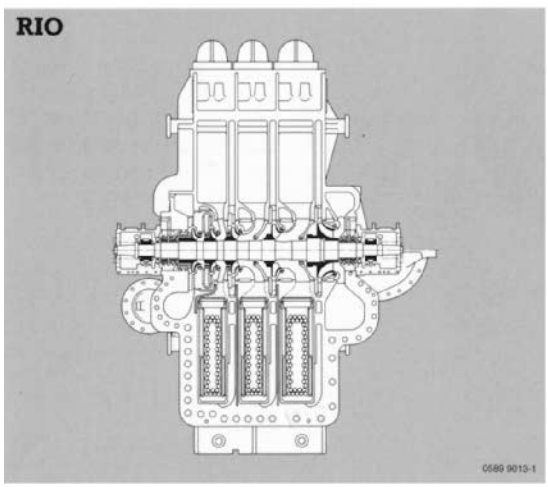
1. *Low power consumption*
   - Intercooling of the gas reduces the inlet temperature into the subsequent stage and therefore its power requirement.
   - Optimization of the distribution of the total cooling surface within the individual cooling stages with respect to heat load, cooling effect, and air-side pressure drop adds to overall efficiency.
   - The short flow path achieved by the single-shaft in-line design with cooler tube bundles integrated in the casing reduces the pressure losses on the gas side as there is no external piping.
   - The staggered high-flow impellers ensure optimal combined performance of all stages, avoiding the lower range of flow coefficients that exhibit a drop in efficiency (Fig. C-194). All impellers are of fully welded or welded and brazed construction (Fig. C-195).
   - In case of dirty cooling water, an automatic cleaning system for the cooler tubes can be installed. This would extend time between overhauls without impairing long-term efficiency.
2. *Resistance against corrosion*
   - Most of the carefully designed flow path is handling hot superheated air; the not-quite-saturated air after the cooler is taken by the shortest way to the next impeller (Fig. C-196).



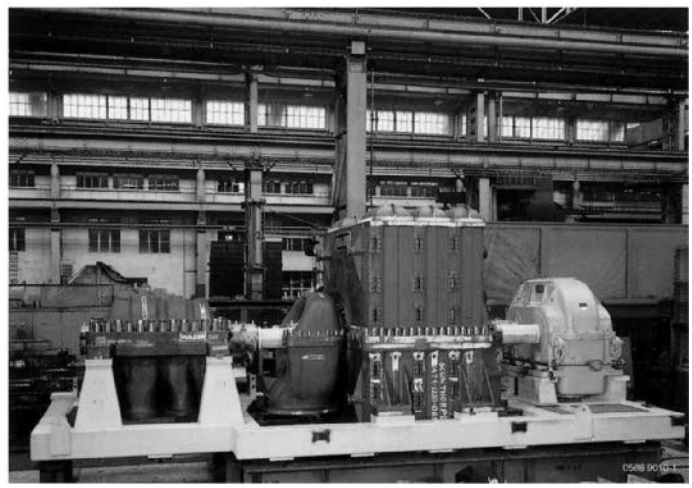
Type RIK
Flow range from 35 000 to 400 000 m³/h.
Nominal discharge pressure up to 13 bar.



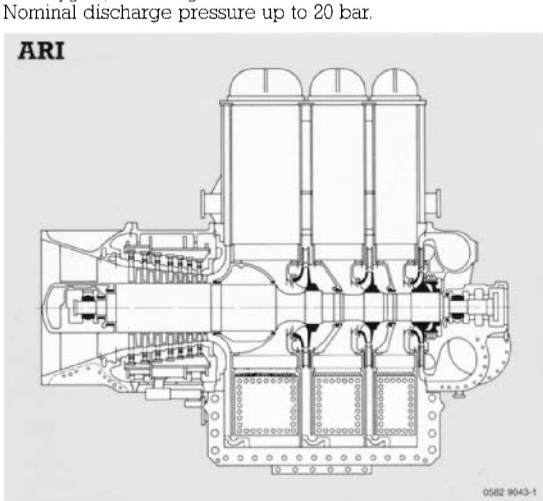
Air compressor, type RIK 71, for an air separation plant in the USA.



Type RIO
For oxygen; flow range from 12 000 m³/h to 90 000 m³/h.
Nominal discharge pressure up to 20 bar.



Oxygen compressor group with LP compressor, type RIO 35, for Europes largest air separation plant



Type ARI
Flow range up to 600 000 m³/h.
Nominal discharge pressure up to 11 bar.



Air compressor, type ARI 56, in a Chinese air separation plant.

FIG. C-193 Operating ranges and applications of isotherm compressors. (Source: Sulzer-Burckhardt.)

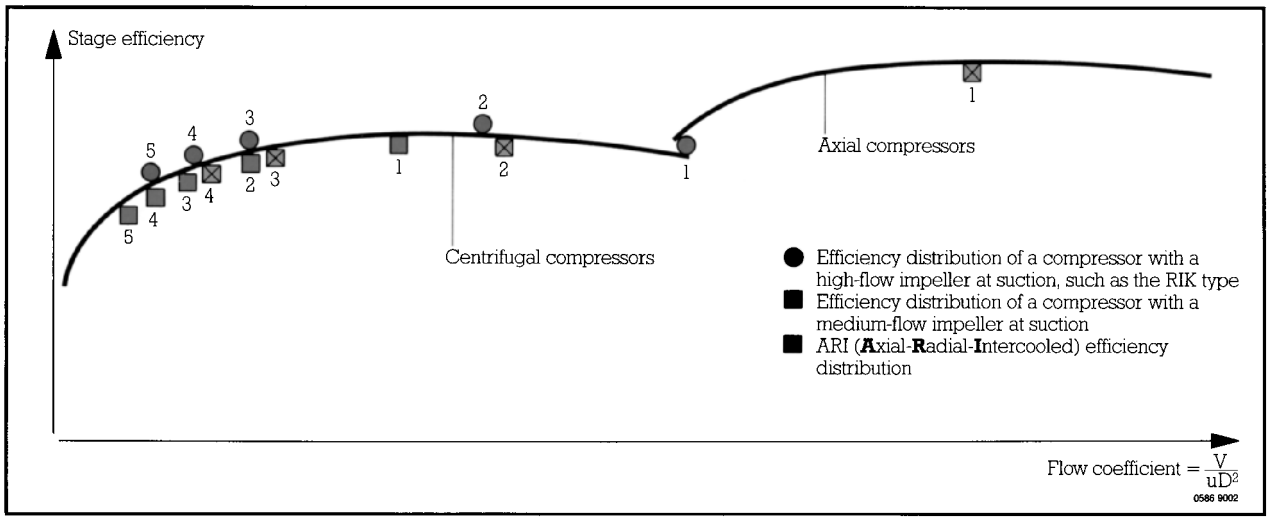


FIG. C-194 Influence of flow coefficient of the first impeller on the efficiency of subsequent stages. (Source: Sulzer-Burckhardt.)



FIG. C-195 Welded impeller of high flow coefficient. (Source: Sulzer-Burckhardt.)

- The vertical position of the built-in lateral cooler tube bundles the inertia-type water separators fitted at the outlet of the coolers (except for the RIO types) to show high separation efficiency enhanced by the effective condensate removal by gravity (Figs. C-197 and C-198). Due to this and the subcooling effect along the tube fins, the air entering the following stage has a mean temperature just slightly above the dew point, which again reduces erosion and corrosion. The condensate is drained by automatic traps.

3. Rotor stability. Radial bearings and special coupling techniques help turbocompressor rotors achieve high stability under all practical operating conditions. This is achieved by the main features illustrated in Table C-14 (Fig. C-199); Figs. C-200 and C-201 show typical rotor assembly layouts.

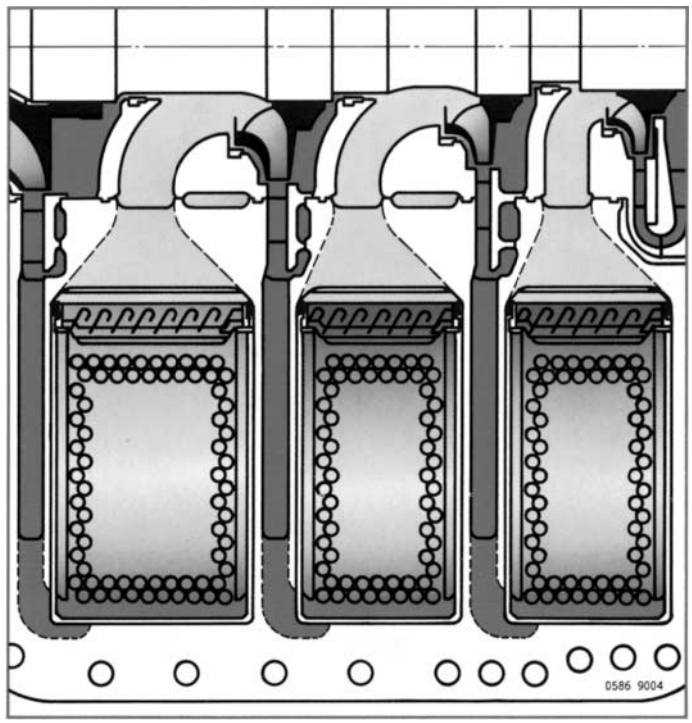


FIG. C-196 Temperature and humidity conditions: hot cool, but well above dew point; cold, but not yet saturated; cold and condensing. (Source: Sulzer-Burckhardt.)

4. Low noise level. The radial casing with the built-in coolers has an attenuating effect on the noise generated by the active high-velocity parts embedded in this compact outer package. The same applies to the double-casing axial part of the ARI series. The noise level is therefore lower than that of a centrifugal compressor with separate external coolers and the necessary interconnecting piping. In case of severe noise level restrictions, a noise hood covers compressor and gears.

5. Simple shaft-string configuration. The single-shaft in-line concept allows a simple configuration of a complete motor or steam turbine driven compressor train with the least number of shafts and bearings. See Figs. C-202 and C-203. The common choices generally are:

a. Steam tubine direct drive with one common axial thrust bearing in the steam turbine and solid coupling with flexible intermediate shaft between turbine and compressor. Four journal bearings. Axial thrust compensated. No additional load on thrust bearing due to torque lock. Standard for ARI and semipackaged RIK types.
b. Steam turbine direct drive with individual thrust bearings. Four journal bearings. Axial thrust not compensated. Additional load on thrust bearing due to torque lock caused by thermal expansion of shafts taken up by conventional gear coupling. Depending on magnitude of transient thermal expansion of compressor and turbine rotor, a diaphragm-type coupling can be used, reducing the additional axial thrust and requiring no lubrication. Alternative for ARI and semipackaged RIK types.

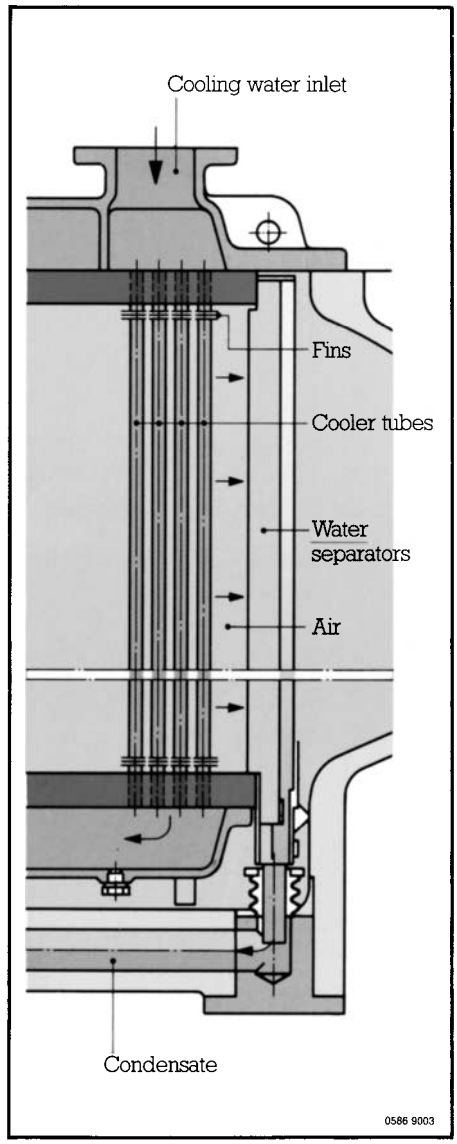


FIG. C-197 Intercooler outlet with water separators. (Source: Sulzer-Burckhardt.)

c. Motor drive with speed increasing gear of conventional design. Axial thrust bearing on compressor and gearwheel shaft. Eight journal bearings including gear and motor. Solid coupling and flexible shaft between compressor and gear; no torque lock. Solution suitable for high-power, low-speed compressors. Preferred alternative for the large ARI types.
d. For higher speeds with motor drive, the OEM applies a solution described in Figs. C-204, C-205A, and C-205B.

Solid couplings with thrust collar gears. Particularly for electric motor driven high-speed compressors of medium to high power and/or pressure, it is normal practice to make extensive use of solid couplings allowing the use of only one axial thrust bearing for single- or multiple-casing arrangements.

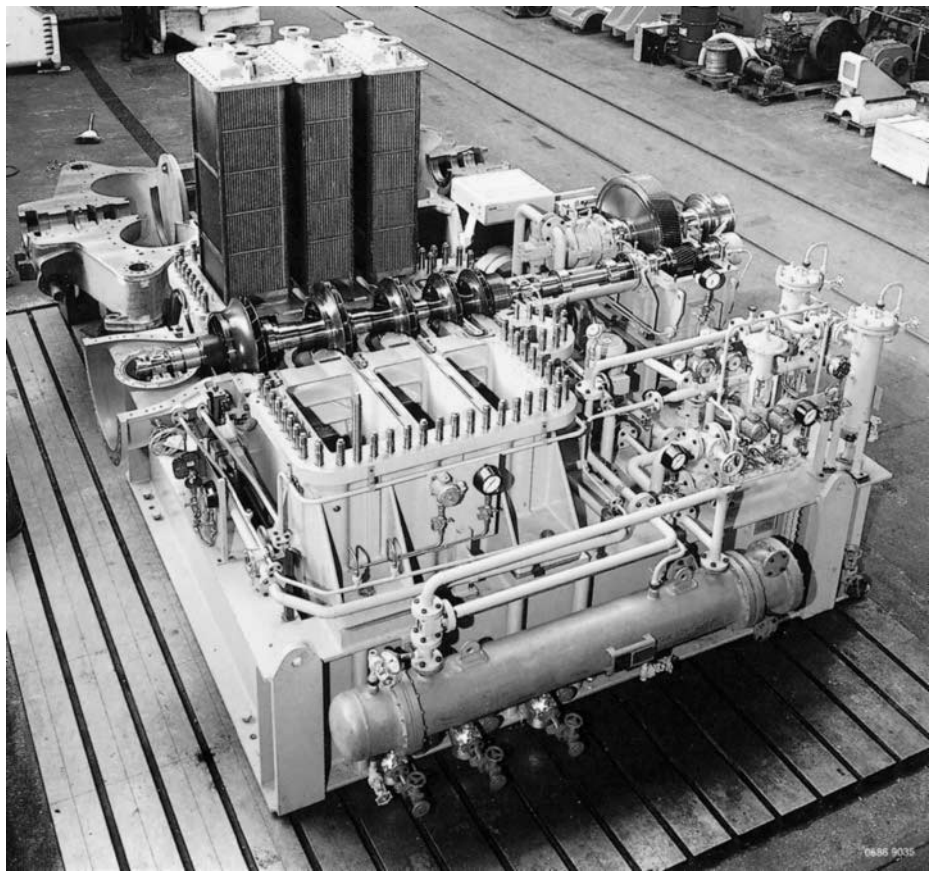


FIG. C-198 RIK model with intercooler tube bundles in the casing bottom half. The vertical water separators at the cooler outlet ensure effective drainage of the condensate. (Source: Sulzer-Burckhardt.)

TABLE C-14 Design Features Illustrated in Figs. C-199 through C-201

Design	Advantages
1 Solid, sturdy rotor with shrunk-on dummy piston	Minimum sensitivity to critical speeds and unbalance due to higher rotor stability; reduction of rotor internal damping
2 Shrink fit secured by symmetrically arranged radial dowels for impellers	No need for keys and distance bushings; fixation ensures concentricity and perfect balance under extreme operating conditions; allows larger shaft diameters; reduces stress on shaft and impeller
3 No shaft sleeves between stages	Reduces rotor hysteresis and increases running stability
4 Labyrinths always on the rotating element. Stainless steel strips caulked into the shaft and impeller grooves	No distortion of rotor due to local heating up in case of rubbing; labyrinths can be refitted easily
5 Nickel plating or other coating of shaft portions exposed to corrosion, if necessary	Plating instead of shaft sleeves is a more direct protection; allows larger shaft diameters
6 Tilting-pad radial bearings for higher speeds	Improves running stability; no oil-whip; higher external damping
7 Solid coupling, tightly bolted to flexible intermediate shaft	Improves reliability due to elimination of high-speed thrust bearing and toothed-type couplings; no torque lock thrust on high-speed thrust bearing
8 High-flow impeller at suction	Improves overall efficiency

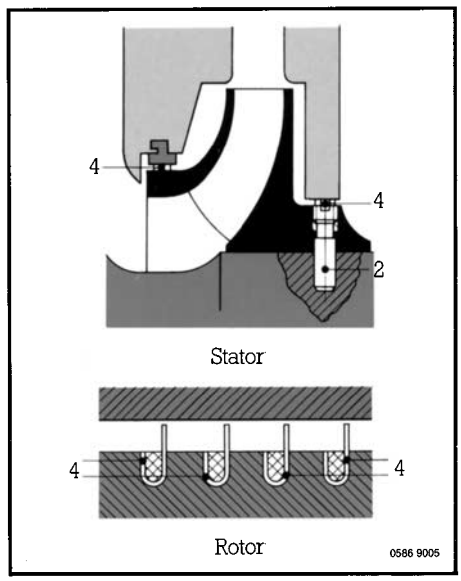


FIG. C-199 Design principles of isotherm turbocompressor rotors. (Match numbers on the figure with features in Table C-14.) (Source: Sulzer-Burckhardt.)

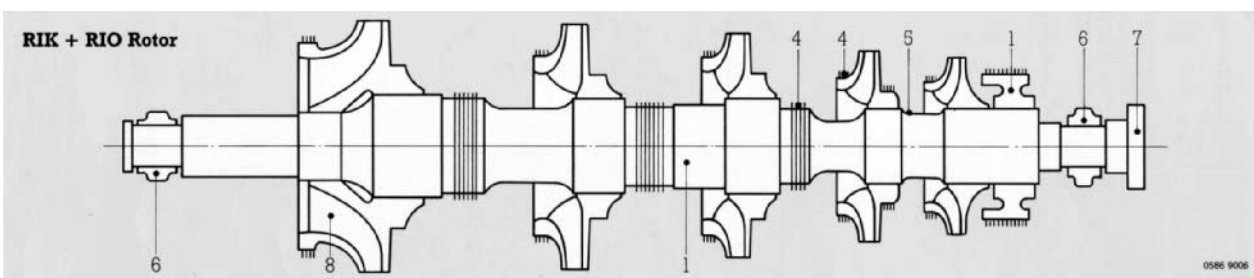


FIG. C-200 Typical rotor assembly layout: RIK and RIO models. (Source: Sulzer-Burckhardt.)

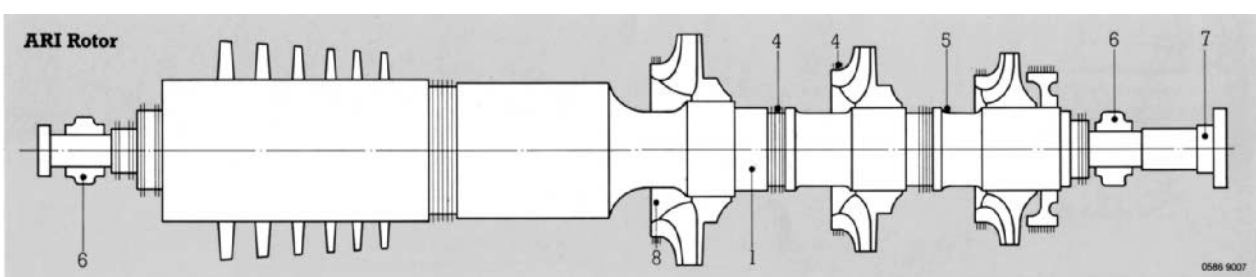


FIG. C-201 Typical rotor assembly layout: ARI model. (Source: Sulzer-Burckhardt.)

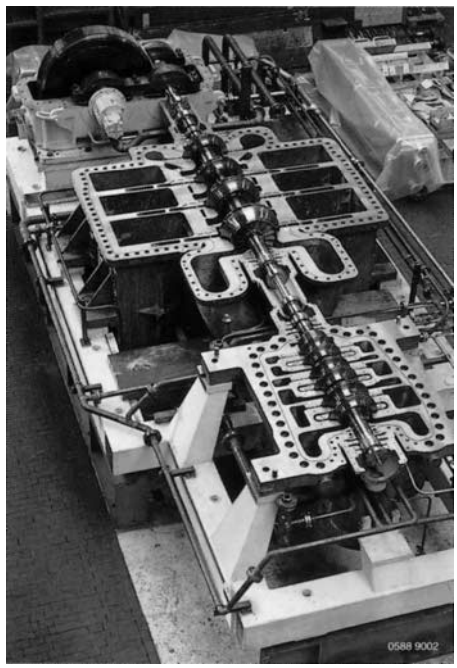


FIG. C-202 Typical shaft-string configuration of a motor-driven isotherm compressor with booster. Axial thrust transmission according to Figs. C-204, C-205A, and C-205B with one single thrust bearing on the low-speed side of main gear. (Source: Sulzer-Burckhardt.)

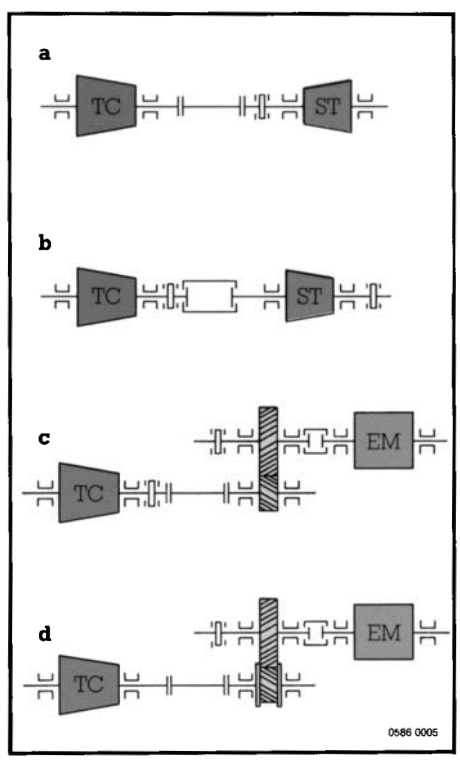


FIG. C-203 Shaft-string configurations. (Source: Sulzer-Burckhardt.)

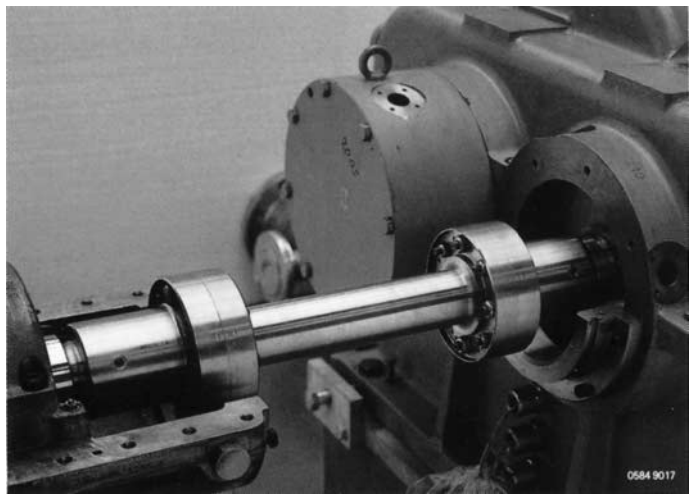


FIG. C-204 The solid quill-shaft coupling conforms to API 671 standard and consists of the quill shaft and the two hubs hydraulically fitted onto the shaft ends of the connected machines. On each coupling side, an equal number of tie bolts for axial fixation and tapered dowel pins for torque transmission and centering ensure a clearly defined connection. Balancing as a complete assembled unit and correlative marking enable removal and remounting of this intermediate shaft with the connected rotors remaining in place, without affecting the balancing quality and vibration behavior of the complete string. (Source: Sulzer-Burckhardt.)

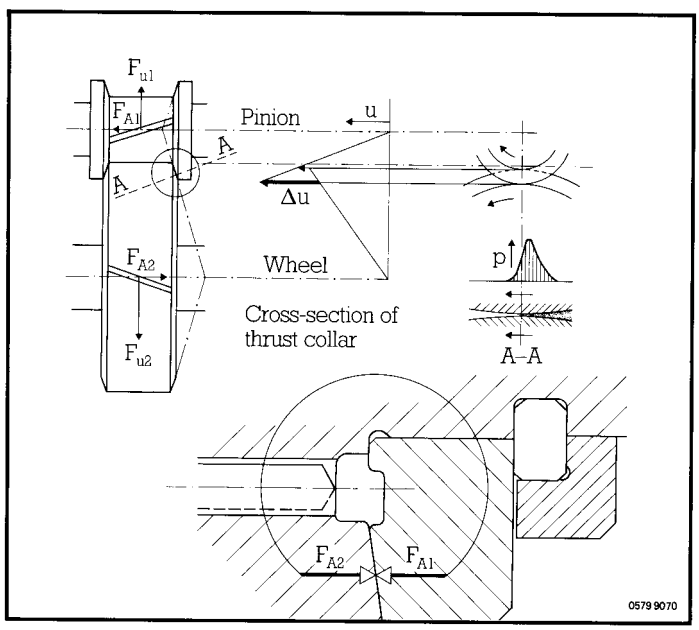


FIG. C-205A Method of axial thrust transfer in a single helical gear with thrust collar. F_u = peripheral force, F_A = axial force, u = peripheral speed, p = pressure. (Source: Sulzer-Burckhardt.)

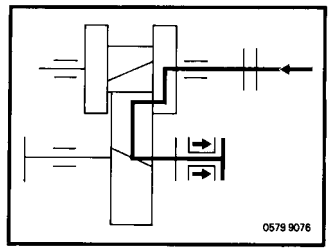


FIG. C-205B Transfer of external forces. (Source: Sulzer-Burckhardt.)

An intermediate shaft, flexible enough to allow for considerable misalignment, is inserted between the two shaft ends of the machines to be coupled together (Fig. C-204). In case of motor-driven units, the normal technique is to use single helical gears provided with thrust collars on the pinion shaft, as shown in Figs. C-205A and C-205B. The thrust collars not only neutralize the axial thrust created by the meshing of the teeth cut at an angle to the axis of the shaft, but also transmit the residual axial thrust of the high-speed rotor train to the thrust bearing on the low-speed wheel shaft.

Good gear meshing requires parallelity of gear and pinion shaft and automatically ensures parallelity of the contact surfaces of thrust collar and wheel rim. The slight tapering of the thrust collars is responsible for the formation of a wedge-type oil film creating a pressure zone spread out on an enlarged surface with a pressure distribution very similar to that of a standard-oil-lubricated journal bearing.

The relative motion between the two contact surfaces of the thrust collar system is a combination of rolling and sliding and takes place near the pitch circle diameter, resulting in a very small relative velocity. The thrust transmission is therefore effected with almost no mechanical losses. The considerably reduced losses of the single thrust bearing on the low-speed shaft as compared with the high losses of individual thrust bearings on the high-speed train lead to a substantial power saving. Moreover, this low-speed bearing can be more robustly dimensioned to provide a much higher overload capacity.

This coupling arrangement avoids heavy overhung gear couplings that are usually responsible for not clearly defined lower critical speeds and for the phenomena of torque lock leading to additional loading of the axial thrust bearing. The resulting axial friction forces can become quite substantial if insufficient attention is given to the cleanliness of the lubricating oil. This arrangement is, therefore, the preferred solution. Its strict application is clearly visible on the air compressor train (Fig. C-202).

6. Design features for erection on site and dismantling for inspection include:
- Package construction
- One single horizontal plane of the axis
- Vertical cooler bundles easily withdrawable
- No heavy and cumbersome crossover piping between compressor and external intercoolers

7. Reduced maintenance, because all components are easily accessible

8. Minimum space requirement through compact single-shaft design with integrated coolers. Low elevation of operating floor for ARI types; skid-mounted

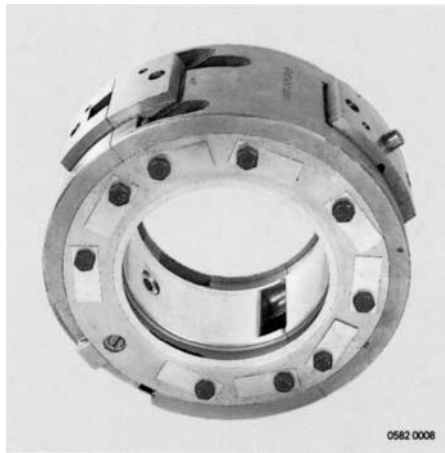


FIG. C-206 Two-lobe journal bearing. (Source: Sulzer-Burckhardt.)

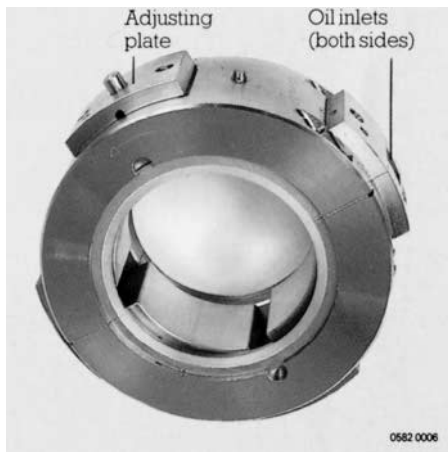


FIG. C-207 Multisegment journal bearing with four tilting pads. (Source: Sulzer-Burckhardt.)

single-life package with integrated gear and lube oil system for RIK and RIO types.

9. High reliability using generic designs

Journal and axial bearings

- *Two-lobe journal bearings* are used on the larger frame sizes of the ARI series running at a moderate speed (Fig. C-206).
- *Tilting-pad journal bearings* are incorporated in the RIK and RIO series operating in a higher-speed range. They contribute to the high rotor stability at high rotational speeds (Fig. C-207).

The horizontally split journal bearings are white-metal-lined and forced-feed-lubricated. Adjusting plates with a slight curvature in axial direction allow the bearings to be set accurately on erection. Shims placed between the plates and the bearing shell make corrective realignment easy. Thermoelement connections for white metal temperature measurement are fitted.

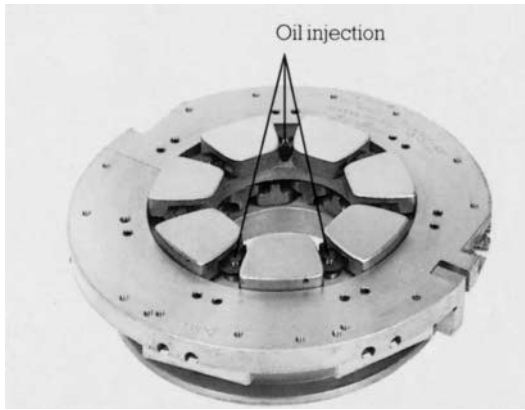


FIG. C-208 Kingsbury-type axial thrust bearing with self-equalized pads with directed lubrication. (Source: Sulzer-Burckhardt.)

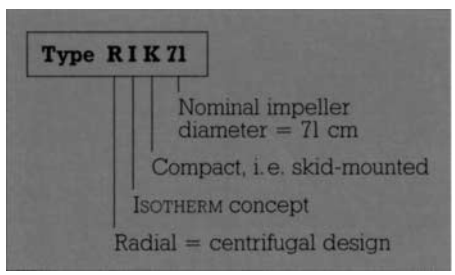


FIG. C-209 RIK series designation example: five centrifugal stages. (Source: Sulzer-Burckhardt.)

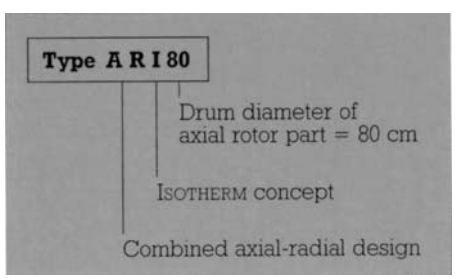


FIG. C-210 ARI series designation example: five axial and three centrifugal stages. (Source: Sulzer-Burckhardt.)

- *The axial thrust bearing* is normally located on the low-speed shaft of the gear. In multicasing arrangements with no gears it is normally located on the intermediate shaft. The thrust bearing is fitted with a load equalizing system. The pads are individually lubricated (Fig. C-208).

Performance data RIK and ARI. See Figs. C-209 and C-210 for type designation examples.

Figures C-211 through C-213 allow selection of the:

Compressor size	Nominal diameter	D (cm)
	Power input	P (kW)

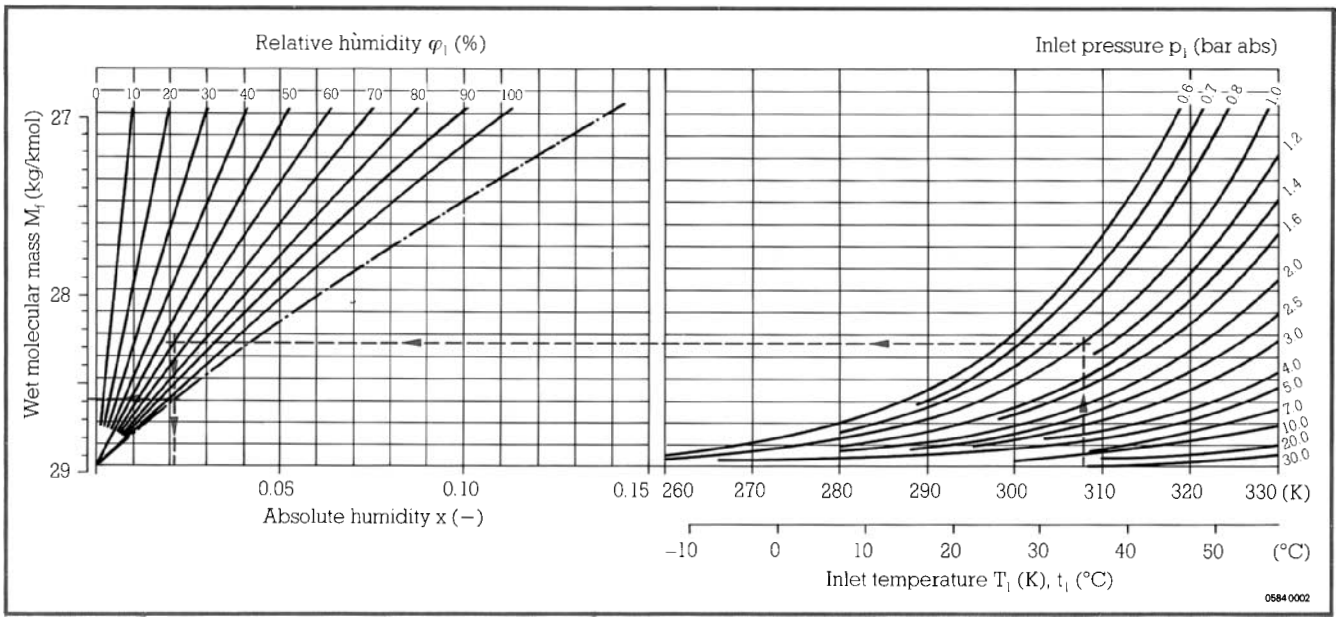


FIG. C-211 Determination of the absolute humidity *x* and the molecular mass M_f of the wet air. (Source: Sulzer-Burckhardt.)

Operating conditions	Mass flow	$\dot{m}$ (kg/s)
	Suction pressure	p_1 (bar abs)
	Suction temperature	T_1 (K)/t_1 (°C)
	Relative humidity of the air or gas	φ_1 (%)
	Discharge pressure	p_2 (bar abs)
	Molecular mass	M (kg/mol)

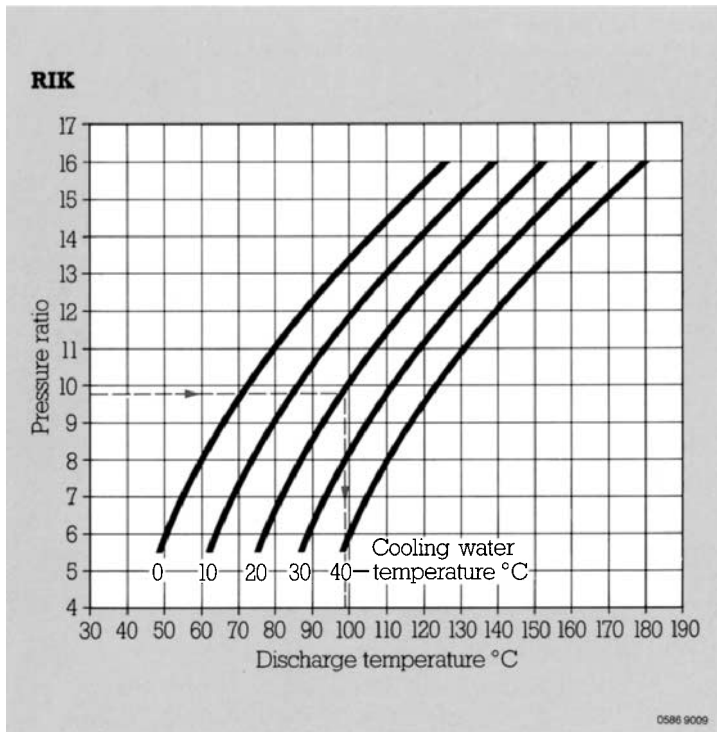
The following factors and symbols are also used for the calculation:

	Suction volume (actual)	$\dot{V}_1$ (m³/h)
	Absolute humidity	x (–)
	Isothermal efficiency	η_{iso} (%)
Indices	Suction branch	1
	Discharge branch	2
	Dry	t
	Wet	f

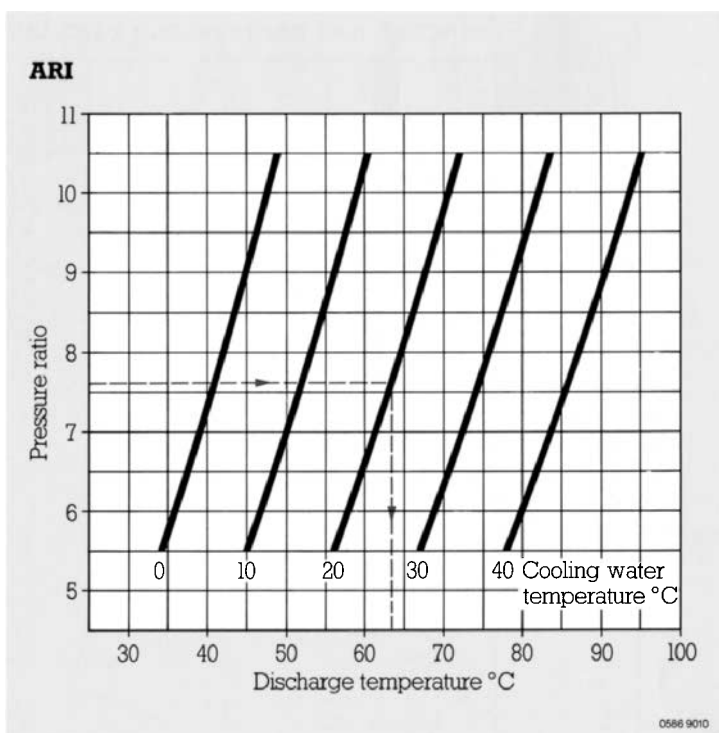
See also Table C-15 for how to specify an isotherm compressor.

In Figs. C-214 through C-216:

- NP = reference point (100 percent) = design point
- α = angular position of the inlet guide vanes (RIK models) or the adjustable stator blades (ARI models)
- Valid for air at constant inlet data.
- Depending on the specific process requirements, such as higher overload capacity, a certain pressure rise to surge, maximum efficiency at design point or rather at a certain part load, the process design point NP may be placed differently in the characteristic curve.



(A)



(B)

FIG. C-212 Determination of the discharge temperature (A) for RIK bodies, (B) for ARI bodies. (Source: Sulzer-Burckhardt.)

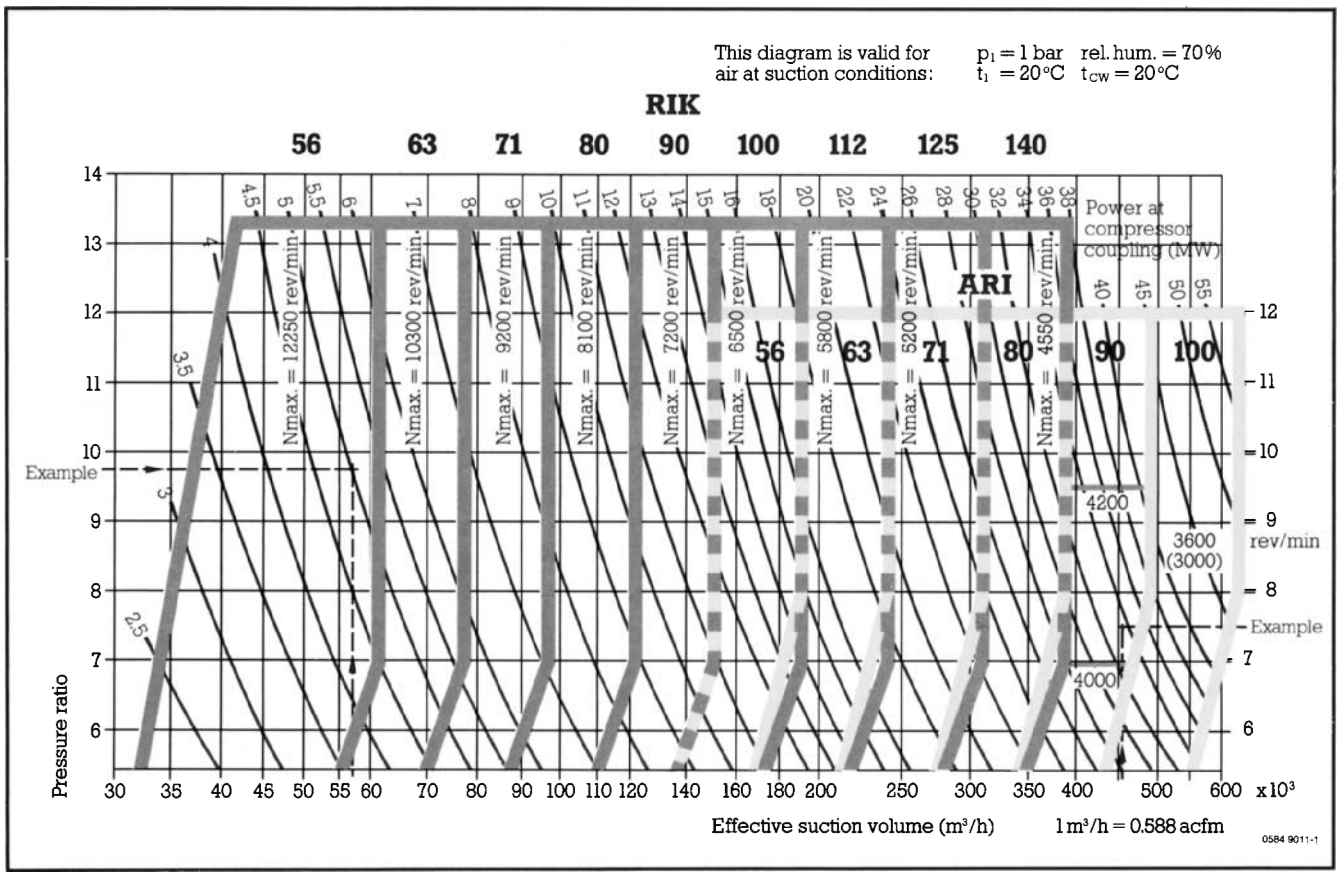


FIG. C-213 Compressor selection diagram. (Source: Sulzer-Burckhardt.)

Design features (see also Table C-16)

Skid-mounted single-lift concept. The various frame sizes are skid-mounted units with built-in intercoolers and integrated lube oil system. The erection on site includes placing the skid on a simple foundation slab, alignments with the driver, and connecting gas and cooling-water piping (Fig. C-217B).

Alternatives to the standard motor-driven concept are possible. For example:

- Compressor directly coupled to the steam turbine driver or expander
- A booster coupled to the compressor, with or without intermediate gear
- Separate freestanding lube oil system
- Suction nozzle facing downward

Casing. See Fig. C-217 for the labeling of internals.

The horizontally split casing contains the five centrifugal stages and three pairs of vertical intercoolers (Fig. C-218). The axial inlet ensures ideal flow conditions through the inlet guide vanes into the first stage. The bearings can be inspected without having to disconnect the gas or oil piping or to disturb the casing top half. The flow passages to and from the coolers are the result of exhaustive model tests and ensures that the gas flow in each stage is equally distributed between the parallel cooler elements (Fig. C-219).

TABLE C-15 Selection and Sample Performance Calculation of an Isotherm Compressor

	Example 1 Type RIK	Example 2 Type ARI
Given		
Mass flow (dry)	$\dot{m}_t$ = 17.29 kg/s	$\dot{m}_t$ = 136.48 kg/s
Suction pressure	p_1 = 1 bar	p_1 = 1 bar
Suction temperature	T_1 = 308 K; t_1 = 35°C	T_1 = 308 K; t_1 = 35°C
Relative humidity	φ_1 = 60%	φ_1 = 60%
Discharge pressure	p_2 = 9.8 bar	p_2 = 7.6 bar
Cooling-water temperature	t_w = 20°C	t_w = 20°C
Dry molecular mass	M_t = 28.96 kg/kmol	M_t = 28.96 kg/kmol
Calculation instructions		
1 Determination of the absolute humidity (x) and the wet molecular mass using Fig. C-211	x = 0.021 M_f = 28.58 kg/kmol	x = 0.021 M_f = 28.58 kg/kmol
Wet gas constant $R_f = \frac{8315}{M_f}$	R_f = 290.94 J/kgK	R_f = 290.94 J/kgK
2 Calculation of the wet mass flow $\dot{m}_f = \dot{m}_t\,(1 + x)$	$\dot{m}_f = 17.29 \cdot 1.021$ = 17.65 kg/s	$\dot{m}_f = 136.48 \cdot 1.021$ = 139.35 kg/s
3 Determination of the actual suction Volume $\dot{V}_1 = \frac{m_f \cdot R_f \cdot T_1 \cdot 3600}{P_1 \cdot 10^5}(\mathrm{m^3/h})$	$\dot{V}_1$ = 56,950 m³/h	$\dot{V}_1$ = 449,520 m³/h
4 Determination of the discharge temperature t_2 with Fig. C-212	t_2 = 98°C	t_2 = 63°C
5 Selection of the compressor frame size and power input P with Fig. C-213	PIK 56 P = 5.0 MW	ARI 90 P = 20.3 MW

Conversion factors

1,000 Nm³/h (1.013 bar, 273 K, dry) = 0.3592 kg/s
1,000 scfm (14.7 psia, 60 F, dry) = 0.5774 kg/s
1,000 scfm (14.7 psia, 70 F, dry) = 0.5665 kg/s
1 m³/h = 0.5886 cfm/1 bar = 14.5 psi

For the hydrostatic tests the casing is divided into several chambers and submitted to a water pressure of 1.5 times the maximum possible operating pressure of the corresponding compartment.

Seals. The shaft and interstage seals are of the labyrinth type. The stainless steel strips are fixed in grooves in the rotating parts (shaft, impeller hub, and cover disc) and have a very small radial clearance against stationary plastic ring segments in the corresponding partition walls. Internal and external leaks are thus kept to a minimum (Fig. C-199).

The axial thrust of the impellers is almost entirely compensated by a balance piston at the discharge end of the compressor. The piston is provided with labyrinth strips rotating against a white-metal–lined steel ring.

With the in-line arrangement of the impellers the resulting total axial thrust of the rotor is, contrary to a back-to-back arrangement, always acting in the same direction and of the same order of magnitude under all operating conditions (normal, reduced load, surge, and rotating stall). The balance piston is dimensioned in such a way that the compensated residual thrust is reduced to a minimum, but still always acting in the same direction. With this method the axial thrust bearing need not be oversized, and the bearing losses are reduced accordingly without any risk of overloading it under abnormal operating conditions.

On the suction side, a special sealing system prevents any oil or oil mist of the bearing space seeping into the surrounding suction ducts thus contaminating, for

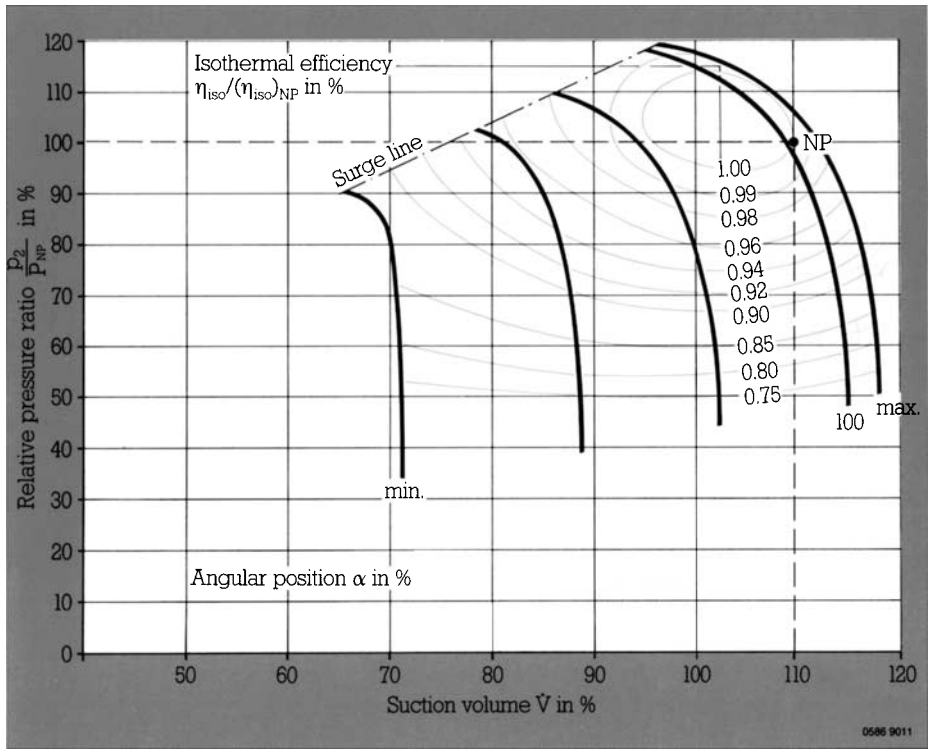


FIG. C-214 RIK compressor with inlet guide vanes and constant speed driver. (Source: Sulzer-Burckhardt.)

example, the air of an oxygen or nitrogen plant (Fig. C-220). The sealing air introduced in the middle of the shaft seal is discharged to atmosphere on the bearing side in order to avoid building up pressure in the confined bearing space connected with the oil tank.

Rotor and impellers (Fig. C-221). The impeller and shaft materials undergo a number of metallurgical tests. Further tests are carried out during manufacture. The finished impellers are then balanced at low speed and subjected to an over-speed test. The assembled rotor is dynamically balanced over the whole speed range up to full speed. Impellers with a medium to high flow coefficient are of the fully welded construction with the blades shaped in three dimensions (see Fig. C-195). Small and narrow impellers are of the combined welded/brazed construction. The sense of rotation of the rotor is clockwise, seen from the suction side.

Inlet guide vanes. To obtain the characteristics as shown in Fig. C-214 with an infinite number of operating points between maximum performance and surge line, inlet guide vanes are fitted ahead of the first stage. They are actuated by a pneumatic servomotor that may be connected to an automatic pressure or flow controller (Fig. C-222). The guide vanes are pivoting in self-lubricating bushes and are connected to an adjusting ring by a maintenance-free linkage system (Fig. C-223). Another link connects the ring with the pneumatic actuator. The absence of a lubricant avoids contamination of the process gas. When starting the compressor, the guide vanes are in an interlocked closed position to reduce the starting torque to a minimum.

Coolers. Three pairs of intercooler tube bundles are mounted in a vertical position on each side of the casing and bolted to the water box. They can expand freely. Round finned tubes ensure excellent heat transfer on the air side. The gap between

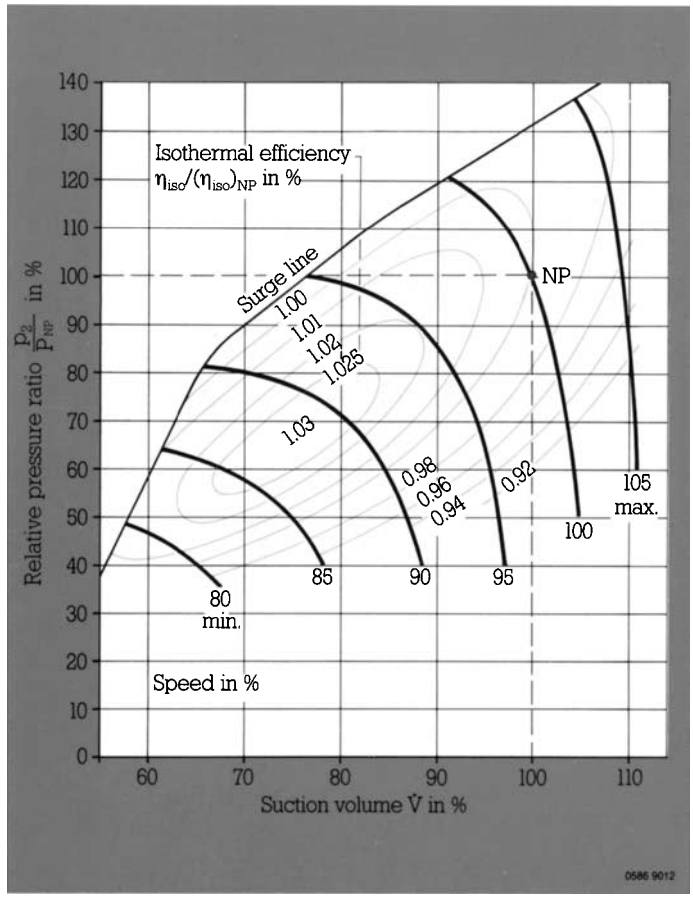


FIG. C-215 RIK compressor without inlet guide vanes, but running at variable speed. (Source: Sulzer-Burckhardt.)

tube bundle and casing at the exit of the cooler is sealed with a rubber membrane to avoid bypassing of uncooled air. See also Figs. C-224 and C-225.

RIK designation technical data and dimensions. See Table C-17 and Figs. C-226 and C-227.

Design features of the RIO isotherm designation. For an RIO series model for compressing oxygen service (see Figs. C-228 and C-229):

- Compact five-stage centrifugal in-line design
- Three pairs of vertical cooler bundles integrated in casing
- Nominal discharge pressure up to 20 bar
- Directly coupled booster compressor available for high discharge pressure

Advantages of the RIO designation. Compressor series has thorough cooling with built-in intercoolers. The compression approaches the ideal of efficient isothermal compression. Result: lowest possible energy consumption and a very compact machine. There are no external coolers, no crossover piping, and no expansion joints. Its simple, lightweight package requires less space, has low overall profile, is easy to erect, and results in minimum installation cost.

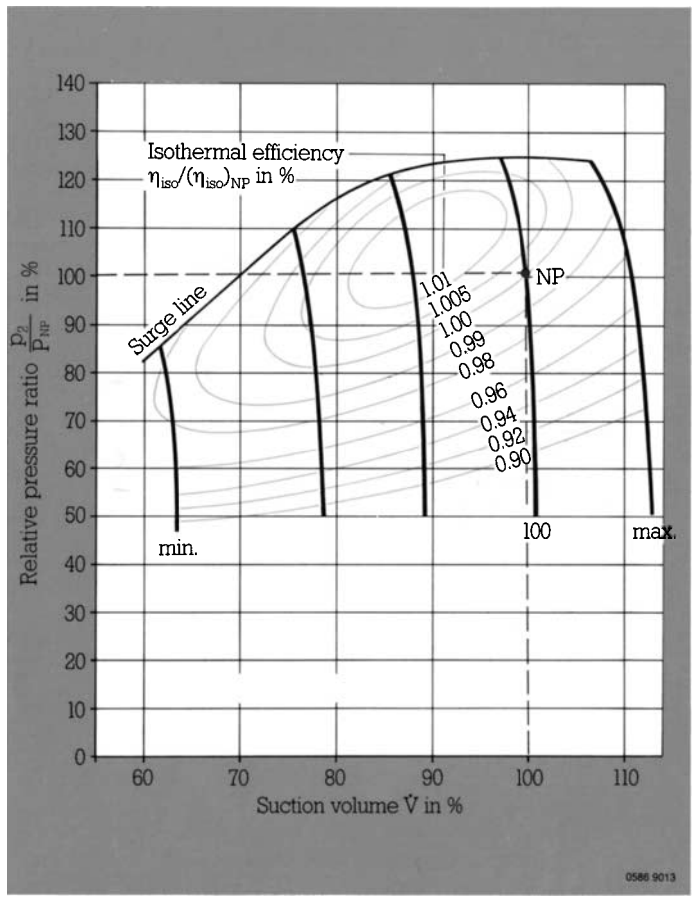


FIG. C-216 ARI compressor with adjustable stator blades and constant speed driver. (Source: Sulzer-Burckhardt.)

Technical data and dimensions. For the RIO designation see Figs. C-230 and C-231. For the ARF designation, design features include:

Axial-radial concept. Where a centrifugal isotherm compressor reaches its economical limits with regard to size, weight, and specific cost, the axial compressor design offers an attractive and technically convincing solution for large volume flows.

For equal aerodynamic loading (Mach number) of the machine and for the same tip diameter of the rotor, an axial stage will handle a volume flow about twice as large as that of a wide centrifugal impeller. If, therefore, the centrifugal section of an isotherm compressor is preceded by an axial booster with a pressure ratio of about 2, the centrifugal section is correspondingly reduced in size and its optimum speed will coincide with that of the axial part. Therefore, the two sections can be combined in one machine with one single rotor running in two bearings only, a design principle similar to that of the industrial single-shaft gas turbine.

Stage and cooler optimization for the predominant pressure ratios for air between 6 and 8 led to a compact axial-centrifugal compressor with six axial and three centrifugal stages and three pairs of intercoolers. This configuration results in an excellent overall isothermal efficiency due to the higher efficiency of the axial part and the high-stage efficiency of the subsequent three wide impellers (see Fig. C-194 and Figs. C-232 and C-233).

TABLE C-16 Materials of Construction

Part	Material	DIN Standard	Comparison ASTM Standard
Casing	Nodular cast iron*	GGG-40*	A 395*
Compressor inlet	Nodular cast iron	GGG-40	A 536
Bearing housing	Nodular cast iron	GGG-40	A 536
Partition walls	Cast iron	GG-20	A 48, Class 30
Diffusers: Discs	Nodular cast iron	GGG-40/1693	A 536
Blades	Carbon steel plates	HI/17155	A 515, Grade 55
Shaft	Low-alloy steel	28 NiCrMoV 8 5	A 470
Impellers	Forged steel	†	†
Dummy piston	Alloy steel	34 CrNiMo 6	AISI 4340
Journal bearings	Steel with white metal	CK 15 + WM	AISI 1015 + WM
Inlet guide vanes	Stainless steel	X 20 Cr 13	
Cooler tubes	Copper	SF-CuF 20 CuNi 10 Fe	
Alternatively:‡	Aluminum brass	CuZn 20 A1 F34/1785	B 111/687
	Copper nickel alloy	CuNi 10 F 29	B 111 C 70600
	Copper nickel alloy	CuNi 30 F 36	B 111 C 71500
Fins‡	Copper	For all tube alternatives	
Tube plates top	Carbon steel plate	HI/17155	A 515, Grade 55
Alternatively:	Muntz metal	CuZn 38 SNAL	B 171 C 36500
Tube plates bottom	Muntz metal	CuZn 38 SNAL	B 171 C 36500
Water separators	Stainless steel	X 5 CrNi 18 9	A 167, Grade 304
Water boxes	Cast iron	GG-20	A 48, Class 30

* Frame sizes RIK 90 and above welded design (carbon steel plate).
† On request, depending on application.
‡ Other alternatives, such as Duplex designs, on request adapted to prevailing cooling-water properties and air contamination.

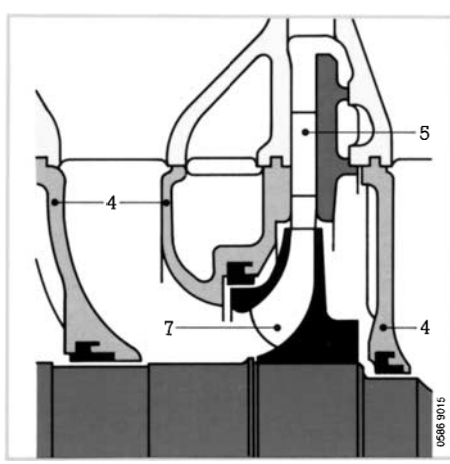


FIG. C-217A Casing internals. Impeller (7), cast iron partition walls (4), vaned diffusors (5). (Source: Sulzer-Burckhardt.)

Casing. The horizontally split casing consists of six major components (Fig. C-233). The cast axial inlet (1) and center part (2) cylinder are flanged to the welded centrifugal casing (3). The inlet casing alone or together with the center part can be lifted for the inspection of their internals while leaving the centrifugal casing in its place. The cast blade carrier (7) and diffuser wall (11) are also bolted to the

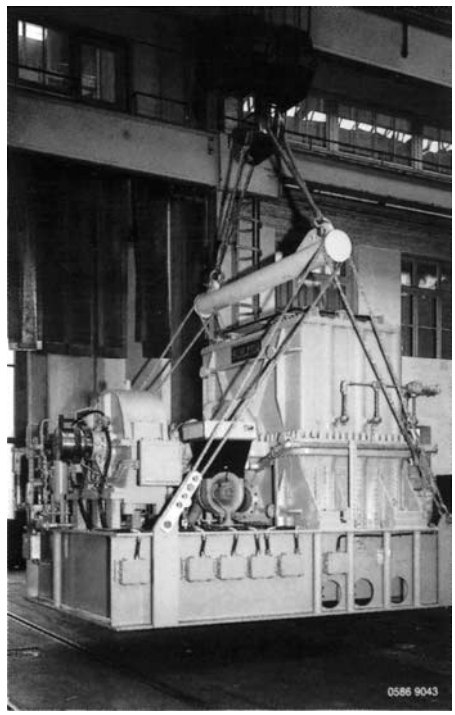


FIG. C-217B RIK skid completely shop erected, as transported to site as a single-lift package. (Source: Sulzer-Burckhardt.)

centrifugal casing. The discharge volute (4) flanged to the centrifugal casing need not be dismantled when lifting the top half of the latter for the purpose of removing the intercooler tube bundles or inspecting the impellers. For all maintenance operations the external pipe connections remain undisturbed.

The bearings are easily accessible by simply lifting the bearing housing top (6) on the discharge side or the top half of inlet casing (1) and bearing housing (5) on the suction side. The oil and sealing-air connections are located in the bottom half of the respective casings, and any inspection or maintenance work leaves them unaffected. The discharge nozzle—forming part of the bottom half discharge volute (4)—is normally pointing downward, but can also be directed horizontally. Two pendulum-type feet and two additional feet attached laterally to the centrifugal cooler casing support the machine on the foundation. See also Fig. C-234.

The flow passages to and from the intercoolers follow the same design principle as for the RIK series. See Figs. C-235 and C-236.

Blade carrier and stator blades. The blade carrier (7) and the short-diffuser wall (11) bolted together are flanged to the cooler casing (3) and can expand freely toward the suction side (Fig. C-233).

The double-casing design with outer casing and blade carrier offers various advantages:

- Rigid casing construction; the clearances in the blade duct are not influenced directly by external pipe forces.
- Simple fitting of the blades and assembly of the casing parts; the top half of the casing can be raised without dismantling the blade-adjusting mechanism.
- Possibility of fitting different blade carriers, for adapting the blade channel and thus the compressor characteristics to greatly changed operating conditions.

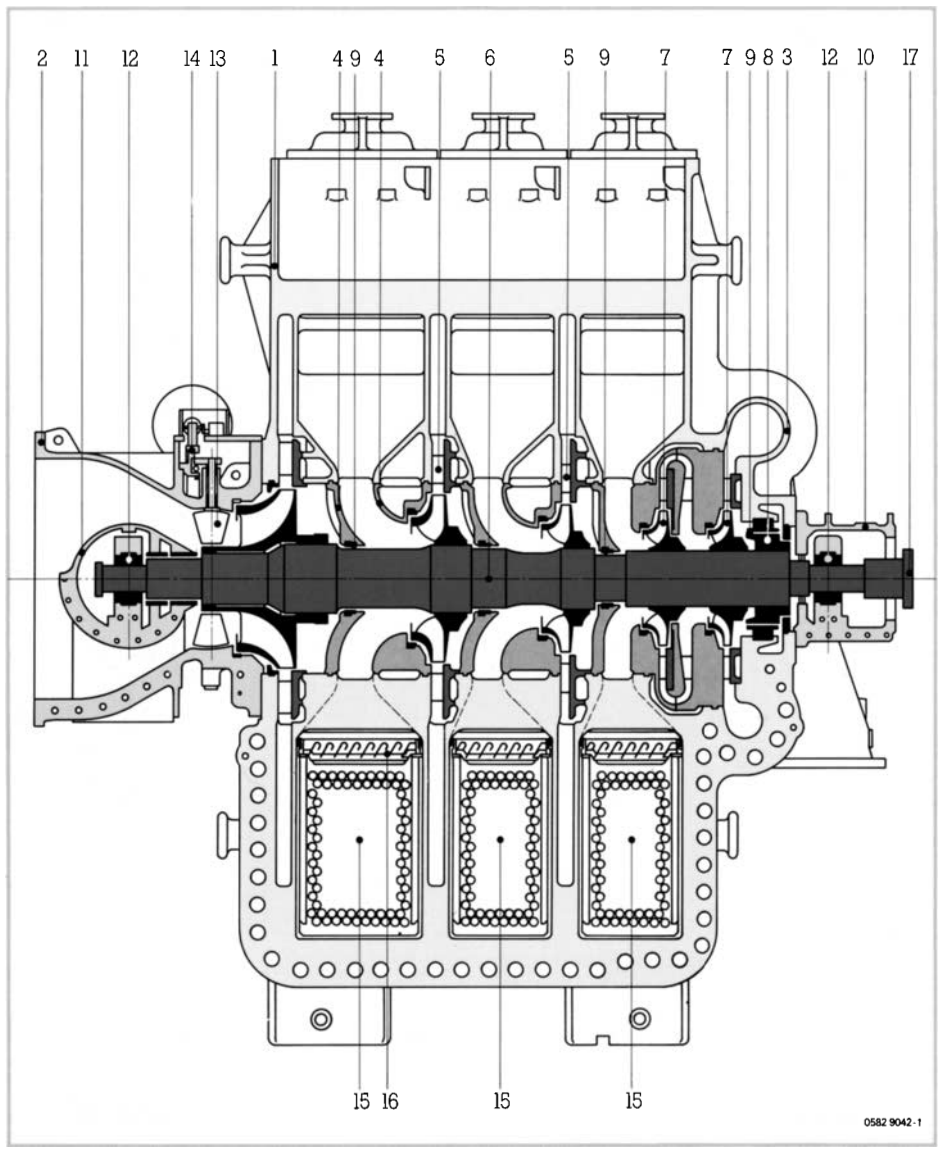


FIG. C-218 Section through an RIK series isotherm compressor (above, vertical section; below, horizontal section). Items 7, 4 and 5 are also illustrated in Fig. C-236. 1, casing; 2, inlet housing; 3, discharge volute; 4, partition walls; 5, diffusors; 6, shaft; 7, impellers; 8, balance piston; 9, shaft seals; 10, discharge-end bearing housing; 11, intake-end bearing housing; 12, journal bearings; 13, inlet guide vanes; 14, vane adjusting mechanism; 15, cooler tube bundle; 16, water separator; 17, coupling flange. (Source: Sulzer-Burckhardt.)

- Optimal protection of the adjusting mechanism in the space between the casing and blade carrier; the space is kept under suction pressure in order to safeguard the adjusting mechanism against condensation and corrosion attack.

Each of the adjustable stator blades (9) is made of one piece with a cylindrical shaft. The latter is seated in a bearing bush in the blade carrier (Figs. C-237 and C-238). The high damping characteristics of this seating arrangement practically excludes the occurrence of dangerous vibration amplitudes associated with the stator blades.

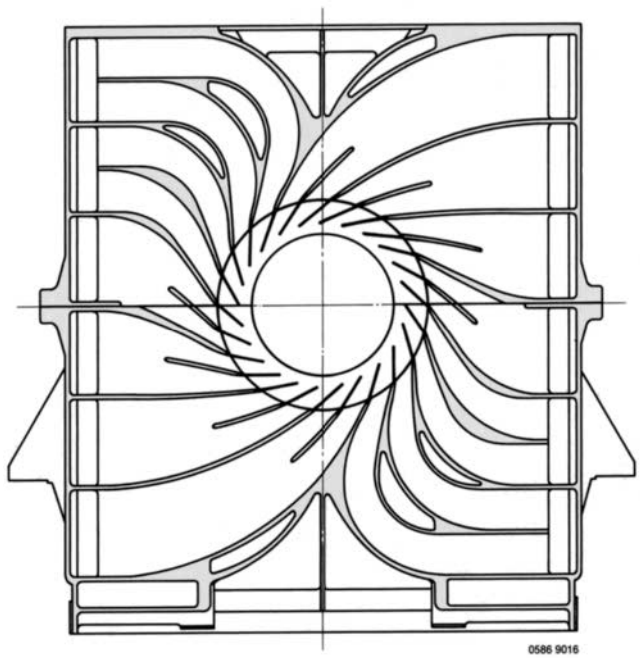


FIG. C-219 Cross-section through diffuser and return channel. (Source: Sulzer-Burckhardt.)

TABLE C-17 Technical Data and Dimensions

	RIK 56	RIK 63	RIK 71	RIK 80	RIK 90	RIK 100	RIK 112	RIK 125	RIK 140
A	4,600	4,480	5,900	6,500	6,700	7,230	8,450	9,300	10,400
B	3,800	4,000	4,200	4,400	4,900	5,270	5,750	6,200	6,500
C	3,400	4,100	3,700	4,000	5,700*	6,600*	7,300*	8,100*	9,000*
D	1,840	2,250	2,040	2,140	3,200	3,350	3,800	4,500	5,000
E	500	560	560	630	630	630	630	710	710
a	800	900	1,000	1,100	1,200	1,400	1,600	1,800	2,000
b	300	350	400	450	500	600	700	800	900
H_1	4,800	5,600	5,400	5,700	7,200	7,750	8,700	9,900	11,000
H_2	6,700	8,200	7,400	7,900	10,100	10,850	12,100	13,700	15,200
H_3	5,300	6,500	6,300	6,700	9,000	9,650	10,800	12,100	13,400
G_1	8.2	11.1	14	17.3	24	29	41	57	80
G_2	1.1	1.4	1.6	2.0	2.3	2.5	3.2	4.5	6.3
G_3	25	33	40	50	69	85	120	165	230
G_4	36	48	53	66	100	120	155	205	270
mr^2	18	27	45	85	150	280	500	880	1,570
Q 4 passes	325	325	430	540	593	680	760	900	1,025
2 passes	690	690	900	1,150	1,186	1,360	1,520	1,800	2,050
F	3,350	3,700	4,500	5,600	5,600	8,000	8,000	10,000	10,000

Technical data, dimensions, and weights
(dimensions in mm, weights G in metric tons)
E = Gear center distance (average)
H_1 = Crane height to lift casing over rotor
H_2 = Crane height to lift cooler tube bundles over casing
H_3 = Crane height to lift casing top half over cooler tube bundles
G_1 = Casing top half
G_2 = Heaviest single cooler tube bundle
G_3 = Bare compressor
G_4 = Complete skid, max.
mr^2 = Compressor rotor mass moment of inertia in $kg\,m^2$, referred to compressor speed, max.
Q = Cooling-water rate of compressor intercoolers in m^3/h
F = Maximum oil filling of oil tank in liters
* incl. servomotor for guide vane drive.

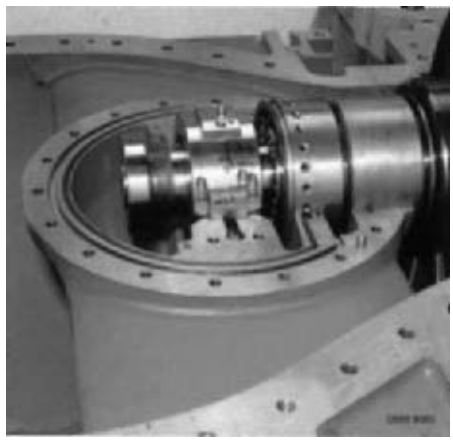


FIG. C-220 Bearing housing suction side with special sealing system. (Source: Sulzer-Burckhardt.)



FIG. C-221 Rotor. (Source: Sulzer-Burckhardt.)

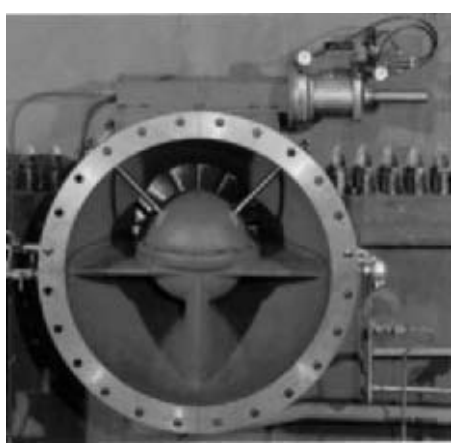


FIG. C-222 Inlet guide vanes with pneumatic actuator. (Source: Sulzer-Burckhardt.)



FIG. C-223 Inlet housing with inlet guide vane linkage system. (Source: Sulzer-Burckhardt.)

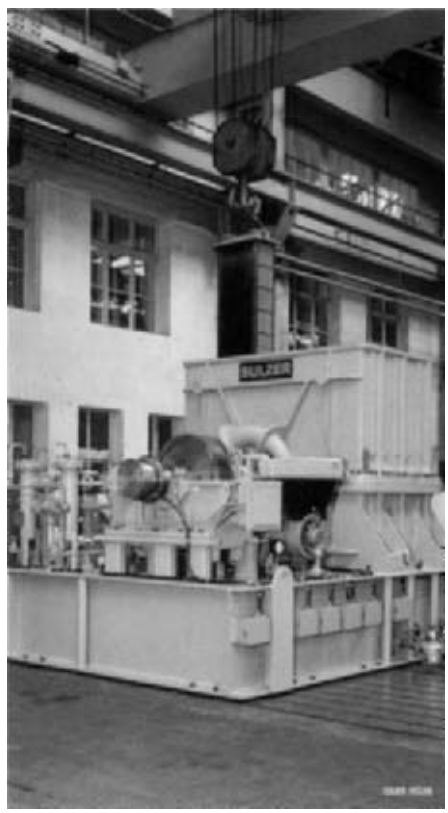


FIG. C-224 For inspection of the intercoolers, each tube bundle can be withdrawn individually. (Source: Sulzer-Burckhardt.)

Stator blade adjusting mechanism. The adjusting mechanism is located in the annular space between casing and blade carrier. It is maintenance-free and does not require any lubrication.

The adjusting mechanism is operated by means of two hydraulic servomotors (10) that are affixed laterally to the bottom-half casing. One of the servomotors is equipped with a positioning transmitter and the second operates hydraulically in parallel (Fig. C-239).

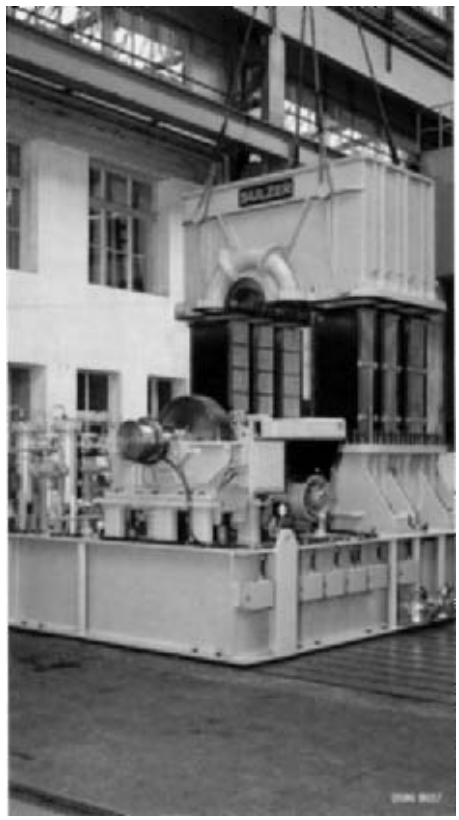


FIG. C-225 For inspection of rotor and casing internals, the intercoolers need not necessarily be withdrawn. (Source: Sulzer-Burckhardt.)

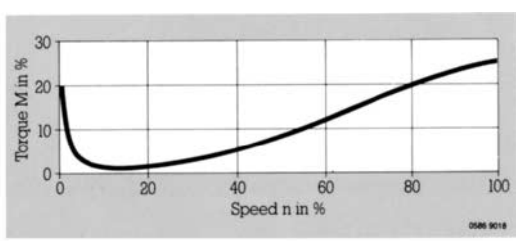


FIG. C-226 Compressor starting torque with closed inlet guide vanes. (Source: Sulzer-Burckhardt.)

The linear movement of the servomotor piston rods is transmitted directly to the adjusting cylinder (8) by way of two ball and socket joints. The adjusting cylinder of welded design can move in the axial direction and is dry-seated. There is no restriction of heat expansion in any direction. U-shaped guide rings are provided on the inner side in which the adjusting levers are engaged. These rings facilitate the assembly of the adjusting cylinder with the stator blade linkages.

The adjusting levers provided on the end of each stator blade shaft are connected to the guide rings of the adjusting cylinder by means of pivoting sliders. The axial movement of the cylinder is converted into a rotating movement of the stator blades (Fig. C-240).

The self-lubricating bearing bushes of the blade shafts are seated in the radial holes of the blade carrier. O-ring packings prevent the ingress of contaminants into the stator blade seating.

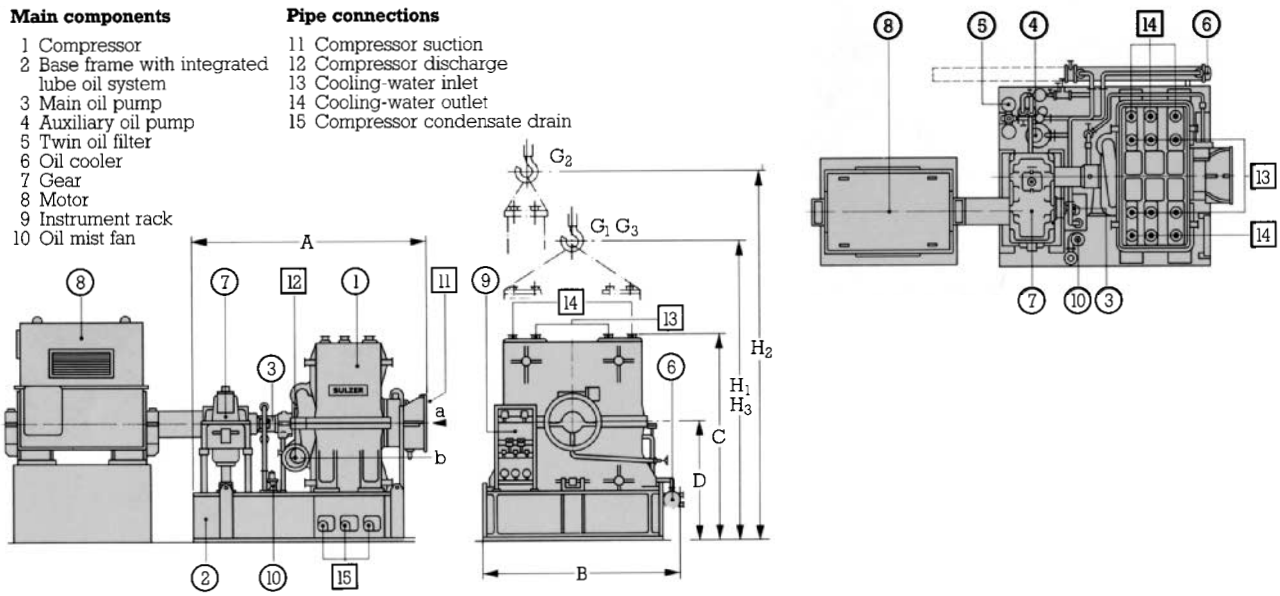


FIG. C-227 Main components and pipe connections (RIK designation). (Source: Sulzer-Burckhardt.)

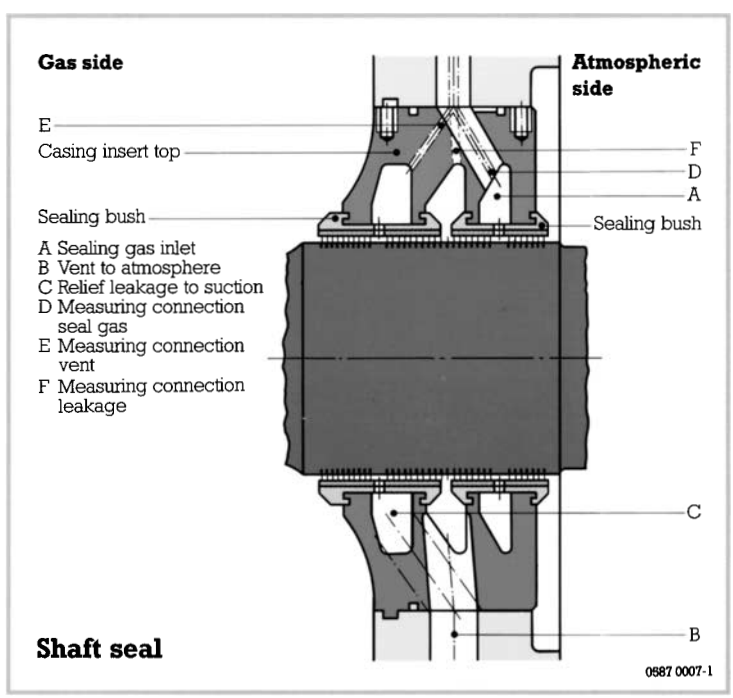


FIG. C-228 Shaft seal layout (RIO designation). (Source: Sulzer-Burckhardt.)

Suction side bearing seal. Double compressed-air sealing on the suction side combined with a double-walled bearing housing (5) that vents to the atmosphere, preventing any sacking in of oil mist in the event of subatmospheric pressure at the machine inlet (Fig. C-241).

Power oil supply. A separate high-pressure control oil unit (Fig. C-242) actuates the hydraulic servomotor of the adjustable axial stator blades. This control oil

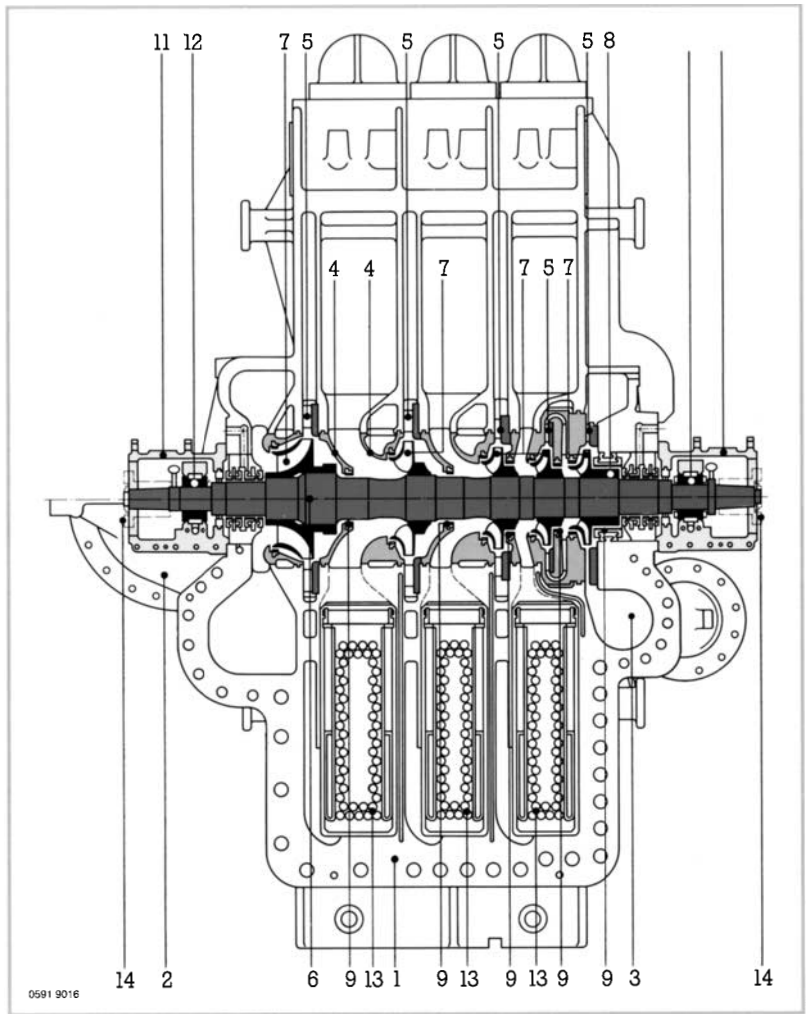
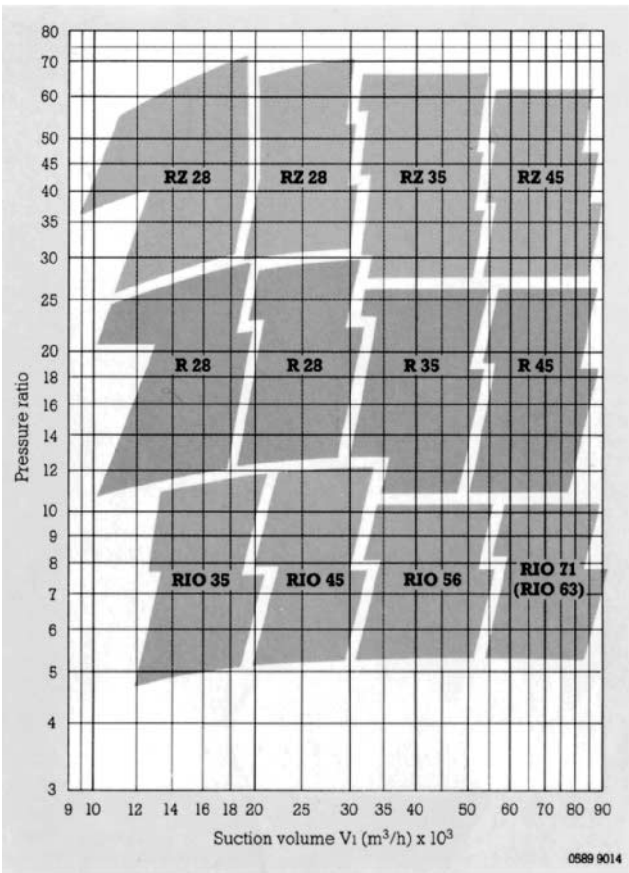


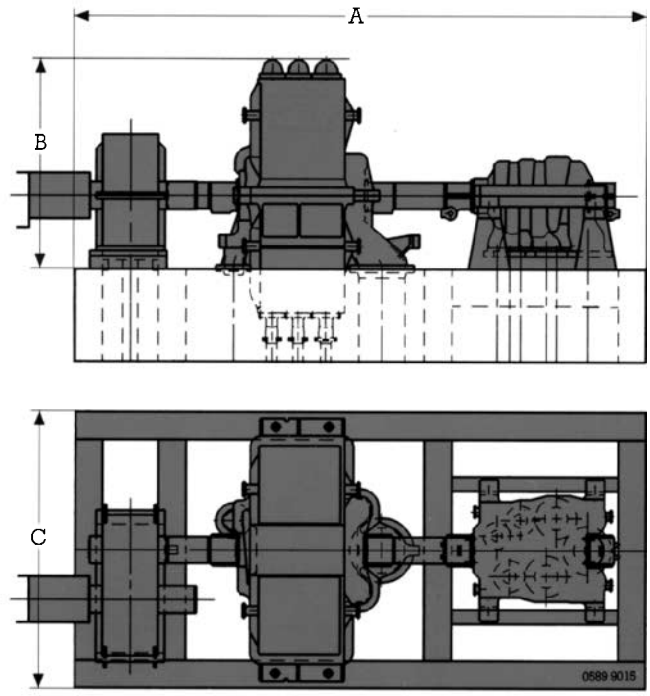
FIG. C-229 Main components (RIO designation). 1, casing; 2, inlet housing; 3, discharge volute; 4, partition walls; 5, diffusers; 6, shaft; 7, impellers; 8, balance piston; 9, seals; 10, discharge-end bearing housing; 11, intake-end bearing housing; 12, journal bearings; 13, cooler tube bundle; 14, coupling flange. (Source: Sulzer-Burckhardt.)

unit comprises an oil tank, two motor-driven pumps, a changeover-type twin oil filter, two bubble accumulators, a regulating valve for constant pressure, and the necessary instrumentation. All components are mounted on a bedplate and piped up accordingly. In case of failure of the control oil pumps the two accumulators will supply enough oil for a quick and safe reaction of the control elements.

Rotor. The basic design features are shown in Fig. C-199 and Table C-14. Both axial and centrifugal parts form a single forged monobloc shaft (see Fig. C-243). The axial blades have rhomboidal fir-tree roots and are firmly braced in an exactly defined position in peripheral grooves of the shaft (Fig. C-238). The three centrifugal impellers of a high flow coefficient, and therefore high efficiency (Fig. C-195), are of the fully welded construction with three-dimensional blades (Fig. C-196). They are balanced at low speed and subjected to an overspeed test. The assembled and bladed rotor is then dynamically balanced over the whole speed range up to full



230



Dimensions for RIO + RZ compressor group

	A	B	C
RIO 35 + RZ 28	6200	2300	3000
RIO 45 + RZ 28	7000	2500	3200
RIO 56 + RZ 35	7200	2600	3200
RIO 63 + RZ 35	7300	2700	3300

231

Technical data (FIG. C-230) and dimensions (FIG. C-231) (RIO designation) of five frame sizes covering flow range from 12,000 to 90,000 m³/h. (Source: Sulzer-Burckhardt.)

speed. The sense of rotation of the rotor is clockwise, seen from the suction side. See also Figs. C-243 and C-244.

Coolers. The three pairs of intercooler tube bundles are mounted in a vertical position on each side of the centrifugal casing and bolted to the lower water box. They can freely expand upward. The upper water box is fixed to the upper tube plate and guided in the top water chamber cover (Fig. C-245). Round finned tubes ensure excellent heat transfer on the air side. The gap between tube bundle and casing at the exit of the cooler is sealed with a rubber membrane to avoid bypassing of uncooled air. The cooling-water connection and condensate drains are located at the bottom.

Integrally geared compressors*

Multistage integrally geared compressors. The design principle of integrally geared centrifugal compressors was established, implemented, and patented by Demag

* Source: Demag Delaval, USA.

Next Page